

CNS2025: Homework 12

Due: 2025-11-26 23:59

Many derivations of optimal conditions in neural systems make use of the Fourier transform, typically in one of its continuous forms. A famous algorithm, Fast Fourier Transform (FFT) is available in almost all numerical libraries. However, the FFT is an implementation of the discrete Fourier transform (DFT). Therefore, care must be taken to map the CFT formulation into a discrete form suitable for the FFT calculation in the library as we have discussed in the lecture.

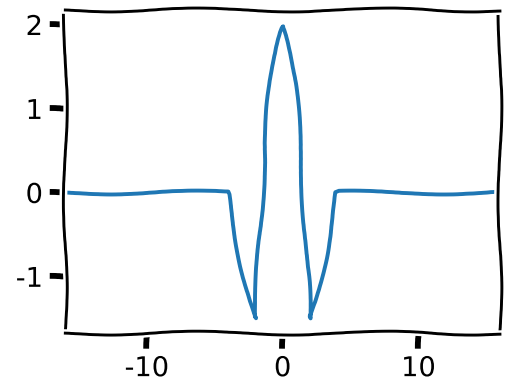
Exercise 1

Let's define the bump function

$$b_m(x, d) = \begin{cases} d^2 - x^2, & \text{if } |x| < |d| \\ 0, & \text{otherwise} \end{cases}.$$

Then, construct the function

$$f_x(x) = b_m(x, 2) - b_m(x, 4)/8.$$



Plot the function for the range, $-L \leq x < L$ with $L = 16$. It should look similar to the illustration to the right above.

Discretize the function for the range $[-L, L)$ into $N = 256$ points. Use the fast Fourier transform provided by `numpy (numpy.fft.fft)` to transform the function into k -space, $[-K, K)$ with $K = \pi N / (2L)$. Plot the Fourier transformed function $f_k(k)$ for the range $-K \leq k < K$. Remember to apply `fftshift` appropriately before and after the transformation so that the resulting f_k is localized around $k = 0$.